Juan R. Perez

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Education

Brown University Spring 2027
BS in Mechanical Engineering, GPA: 3.3/4.0 *Providence, RI*

- **Relevant Coursework:** Statics, Dynamics, Thermodynamics, E&M, Mechanics of Solids and Structures, Fluid Mechanics, Computer-Aided Design

Work Experience

Undergraduate Student Researcher Spring 2026
Wilhelmus Lab – Brown University *Providence, RI*

- Manufacturing biomimetic underwater robots for metachronal propulsion research.
- Manufactured 40+ electromagnetic coils and full robot assemblies (3D printing, soldering, integration) with strict geometric consistency.

Legal Intern Summer 2025
Hanna Perez Personal Injury Attorneys *Paramus, NJ*

- Performed paralegal duties in active personal injury litigation; drafted legal documents, assisted with client meetings and depositions, and optimized office calendar for readability.

Projects

Airfoil Design and Optimization Spring 2026
ENGN 1700: High Reynolds Number Flows – Brown University *Providence, RI*

- Designed and tested optimized airfoil geometries using XFLR5, CFD (MFC/VisIt), and wind tunnel experiments, improving aerodynamic efficiency by 300%+ and achieving $C_L \approx 2.3$.
- Analyzed lift/drag polars, pressure distributions, and vorticity fields to inform geometry modifications (camber, thickness, leading-edge radius) for application-specific performance.

Hydrogen Production Heat Cycle Design Spring 2025
ENGN 0720: Thermodynamics – Brown University *Providence, RI*

- Designed a multi-subsystem hydrogen production process delivering 30 mol/s H_2 at 250 bar; reduced compression energy ∼30% and captured 0.41 MW waste heat, increasing annual revenue by ∼$0.5M.
- Delivered a comprehensive engineering report with quantitative performance analysis and design rationale.

Mechanical Properties & Microstructure Analysis of Structural Materials Fall 2024
ENGN 0410: Materials Science – Brown University *Providence, RI*

- Characterized mechanical properties of 4140 steel, 7075-T6 aluminum, and HDPE using tensile, impact, and hardness tests; generated stress-strain curves to determine elastic modulus, yield strength, and toughness.
- Determined tempered martensite achieved the optimal strength-ductility: 87% of quenched martensite's strength at 40× the ductility.

Human-Centered Chair Design & Fabrication Fall 2023
ENGN 0032: Introduction to Design – Brown University *Providence, RI*

- Built full 3D CAD assembly in Fusion 360 and ran FEA simulations to analyze stress distribution under 750 N and 500 N load cases, iterating geometry to optimize structural integrity.
- Created complete BOM with cost and manufacturability rationale; machined final prototype to GD&T tolerances.
- Prototype withstood 1.5× design load with no visible deformation; earned a class-high top-3 peer evaluation and a 99.6% project grade.

Skills

Design Software: SolidWorks, Fusion 360, SketchUp
Analysis Tools: FEA, CFD, Ansys, Abaqus, VisIt, XFOIL, XFLR5, Instron Equipment, Wind Tunnel Testing, Charpy impact, Rockwell C hardness
Programming/Electronics: Python, C/C++, MATLAB, LTSpice, Arduino
Manufacturing: FDM/SLA 3D Printing, Coil winding, Soldering, Sheet metal, GD&T, BOM